



Petra Michael

Software Developer & Researcher

📍 Jakarta, Indonesia

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 github.com/aimatochysia

PROFESSIONAL SUMMARY

Passionate Software Developer and Researcher from Indonesia with expertise in building cross-platform applications, implementing post-quantum cryptography, and developing data-driven solutions. Experienced in full-stack development with proficiency in Flutter, Python, JavaScript/React, and various backend technologies. Strong focus on security, performance, and user experience. Active open-source contributor with 70+ public repositories showcasing diverse technical projects.

EDUCATION

Bina Nusantara University (BINUS)

Bachelor of Computer Science

Bandung, West Java, Indonesia

Expected 2025

WORK EXPERIENCE

Software Development Engineering Intern

Feb 2025 - Feb 2026

IDEMIA — South Jakarta, Indonesia (On-site)

- Handling internal software development testing for identity and security solutions involving international requirement
- Develop and maintain automated testing frameworks using Jython and Jenkins
- Work with APDU Protocols, EMVs standards, and Javacard technologies for smart card interfaces
- Collaborate with development teams on software testing strategies and software implementation improvement

Research Assistant

Jun 2023 - May 2024

Bina Nusantara University — Bandung, West Java, Indonesia (Hybrid)

- Dataset cleaning for AI training application in fishpond water quality prediction based on image input
- Applied machine learning techniques and Python for data processing and model development
- Contributed to research initiatives combining computer vision with environmental monitoring

Website Maintainer

May 2023 - Apr 2024

Bina Nusantara University — Bandung, West Java, Indonesia (Hybrid)

- Managed and maintained apski.net, a collaboration between universities with School of Business programmes across Indonesia
- Project is tied and supported by the Ministry of Cooperation and Small Businesses of Indonesia (KemenkopUKM)
- Added and fixed web content, ensuring optimal user experience and site functionality

ORGANIZATIONAL EXPERIENCE

Treasurer Nippon Club - BINUS University — Bandung, West Java, Indonesia (Hybrid) - Managed financial operations for future event planning and organizational expansion - Oversaw budgeting, financial accounting, and treasury accounting for non-profit operations	Feb 2024 - Jan 2025
Activist Nippon Club - BINUS University @Bandung, West Java, Indonesia - Led a charity program and oversaw the general expo of the organization - Contributed to event management and social change initiatives	Feb 2023 - Jan 2024
Academic Events Manager HIMTI BINUS University — Bandung, West Java, Indonesia (Remote) - Led a team of education activists to teach programming hard skills - Organized training sessions for School of Computer Science members and the public - Managed operational control and operations for academic events	Feb 2024 - Jan 2025
Activist HIMTI BINUS University — Bandung, West Java, Indonesia (Hybrid) - Taught programming hard skills to School of Computer Science members and the public - Contributed to event management and project management initiatives	Feb 2023 - Jan 2024

TECHNICAL SKILLS

Programming Languages: Python, Dart, JavaScript, TypeScript, Java, Jython, C, C++, Rust, GDScript	Frameworks & Libraries: Flutter, React.js, Next.js, Node.js, Flask, Godot Engine, Chrome Extension APIs
Data & Analysis: Jupyter Notebook, Pandas, NumPy, yfinance, YOLO, CNN	DevOps & Tools: Git, GitHub Actions, Vercel, CI/CD Pipelines

FEATURED PROJECTS

Q-Safe Vault

github.com/aimatochysia/qsafevault

Dart, Flutter, Rust FFI | Post-Quantum Cryptography

Secure, local-first password manager featuring post-quantum cryptography. Implements NIST-standardized Kyber ML-KEM 768 for key encapsulation and Dilithium3 for digital signatures. Features AES-256-GCM encryption, Argon2id key derivation, peer-to-peer sync with end-to-end encryption, and hardware security module support (TPM2, SoftHSM, Secure Enclave).

FicBatch

github.com/aimatochysia/FicBatch

Dart, Flutter | Cross-Platform Mobile & Desktop

Cross-platform Archive of Our Own (AO3) browser and reading application. Features embedded web browsing with custom enhancements, dedicated reader with autosave, local library management, background sync for chapter updates, and reading history tracking. Available on Android and Windows.

Stock Screener (JKSE)

github.com/aimatochysia/stock-screener

Python, React.js, GitHub Actions | Data Pipeline & Web

Automated stock screening system for Jakarta Stock Exchange (~1,000 stocks). Implements distributed data pipeline with daily OHLCV collection, technical analysis (RSI, ATR, SMA, support/resistance), and interactive React frontend. Features automated GitHub Actions workflows for data processing across multiple repositories.

Wallpaper Engine Code Clock

github.com/aimatochysia/Wallpaper-Engine-Code-Clock

HTML, JavaScript, CSS | Desktop Customization

Creative Wallpaper Engine wallpaper displaying a live clock styled as programming language syntax. Supports multiple languages including Python, C, C++, C#, Java, and JavaScript. Published on Steam Workshop.

ADDITIONAL PROJECTS

Maritime Warning System: Illegal ship detection using Global Fishing Watch API

RPS AI: CXR scan image based input Abnormalities classification using YOLO object detection

Pinoted: Note-taking application with sticky notes that stay on screen

Color Picker Extension: Chrome extension for precise color selection with HEX/RGB toggling

Discord RPC: Rich presence tool with dynamic application cycling and GIF support

LANGUAGES

Indonesian: Native

English: Professional Proficiency

INTERESTS

Post-Quantum Cryptography, Financial Technology, Game Development, Open Source Development, Machine Learning, Cross-Platform Application Development, Astrophysics, Material Science, Frontend Design & UX, IoT Projects, 3D Animation & Illustration (Blender3D, Pixel Art)